Development of an Optical Tracking System for Real-time Robotic Applications

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Abstract

Engineering students often need a clear, practical model for writing dissertations; this document provides an overview tailored to final-year mechatronics projects and demonstrates effective use of a LaTeX thesis template. Many student reports lack consistent structure, rigorous scoping, and appropriate use of figures, tables, code listings, and citations, which impedes readability and assessment.

This work contributes an exemplar that explains what each report section should contain, offers typical subheadings, and embeds working examples of template elements to copy and adapt. The document is implemented in XeLaTeX using the `memoir` class with chapters for Introduction, Literature Review, Methodology, Results, Discussion, Conclusions, Recommendations, and Appendices, along with acronyms and bibliography support; the result is a shareable guide that doubles as a starting point for students to produce professional, evidence-based engineering reports.

How to write an abstract in 5 minutes?

Answer the following questions in narrative form with no more than two sentences per question:

- *What is the background to the project?*
- *What is the problem that is being solved?*
- *What is the contribution of the project to this problem?*
- *What is the main conclusion of the project?*

Contents

List of Figures **v**

List of Tables **vi**

1 Introduction **1**
1.1 Background and Motivation 1
1.2 Context and Related Work 1
1.3 Aims, Objectives and Contributions 1
1.4 Project Scope and Limitations 1
1.5 Report Outline 1

2 Literature Review **2**
2.1 Introduction 2
2.2 Overview of Motion Capture 2
2.3 Survey of Optical Motion Capture 2
2.4 Applications of Optical Motion Capture 2
2.5 Limitations of Existing Approaches 2
2.6 Summary 2

3 System Specifications **3**
3.1 Introduction 3
3.2 Overall System Architecture 3
3.3 OCP Subsystem Requirements 3
3.3.1 Performance Requirements 3
3.3.2 Functional Requirements 3
3.4 OMC Subsystem Requirements 3
3.4.1 Performance Requirements 3
3.4.2 Functional Requirements 3
3.5 Constraints and Assumptions 3

4 Optical Motion Capture **4**
4.1 Literature Review 4
4.1.1 Estimation Algorithms 4
4.1.2 Hardware Considerations for Camera-Based Sensing 4
4.2 Camera Selection and System Integration 4
4.3 Theory Development 4
4.3.1 Rigid-body Motion 4

4.3.2 Image Acquisition and Formation 4
4.3.3 Image Processing 5
4.3.4 Triangulation 5
4.3.5 State Estimation 5
4.4 Tracking Algorithm Design 5
4.4.1 Software Architecture 5
4.5 Implementation 5
4.5.1 Calibration Procedure 5
4.5.2 Graphical User Interface 5
4.6 Experimental Testing Setup 5
4.6.1 Physical Configuration 5
4.6.2 Reference System and Ground Truth 5
4.6.3 Test Trajectories 5
4.6.4 Data Acquisition and Processing Pipeline 5
4.7 Results and Discussion 6
4.7.1 Calibration Accuracy 6
4.7.2 Static Localisation Performance 6
4.7.3 Dynamic Tracking Performance 6
4.7.4 Filter Comparison 6
4.7.5 Real-time Performance 6
4.7.6 System Limitations and Failure Analysis 6

5 Optimal Camera Placement 8
5.1 Literature Review 8
5.1.1 Motivation for Optimal Placement 8
5.1.2 Optimisation Approaches for Sensor Placement 8
5.1.3 Modelling the Problem 8
5.1.4 Summary 8
5.2 Theoretical Framework 8
5.2.1 Genetic Algorithm Terminology 8
5.2.2 Encoding Schemes 8
5.2.3 Selection Methods 8
5.2.4 Crossover Techniques 8
5.2.5 Convergence Analysis 8
5.3 Methodology 8
5.3.1 Algorithm Configuration 8
5.3.2 Cost Function 8
5.3.3 Termination Criteria 8
5.4 Results and Discussion 8
5.4.1 Convergence Behaviour 8
5.4.2 Cost Function Evaluation 8
5.4.3 Parameter Sensitivity Analysis 8
5.4.4 Optimal Placement Configurations 8
5.4.5 System Limitations 9

6 Conclusion and Future Work 10
6.1 Summary of Contributions 10

6.2 Key Findings 10
6.3 Limitations 10
6.4 Recommendations for Future Work 10

List of Figures

List of Tables

Introduction 1

1.1 Background and Motivation

1.2 Context and Related Work

1.3 Aims, Objectives and Contributions

1.4 Project Scope and Limitations

1.5 Report Outline

Literature Review 2

2.1 Introduction

2.2 Overview of Motion Capture

2.3 Survey of Optical Motion Capture

2.4 Applications of Optical Motion Capture

2.5 Limitations of Existing Approaches

2.6 Summary

System Specifications 3

3.1 Introduction

3.2 Overall System Architecture

Block Diagram

3.3 OCP Subsystem Requirements

3.3.1 Performance Requirements

3.3.2 Functional Requirements

3.4 OMC Subsystem Requirements

3.4.1 Performance Requirements

3.4.2 Functional Requirements

3.5 Constraints and Assumptions

Optical Motion Capture 4

4.1 Literature Review

4.1.1 Estimation Algorithms

4.1.2 Hardware Considerations for Camera-Based Sensing

4.1.2.1 Camera Capabilities

Resolution, latency, iframe rate, microcontroller...

4.1.2.2 Communication Architectures

Wired vs wireless: PoE, Wi-Fi, Bluetooth Synchronous vs Asynchronous

4.2 Camera Selection and System Integration

4.3 Theory Development

4.3.1 Rigid-body Motion

4.3.1.1 Reference Frames

4.3.1.2 Representing Pose

4.3.2 Image Acquisition and Formation

Computer vision, light and colour, image formation

4.3.2.1 Colour Spaces

4.3.2.2 Camera Model and Projection

4.3.2.3 Camera Calibration

4.3.3 Image Processing

4.3.3.1 Marker Design and Detection

4.3.4 Triangulation

4.3.5 State Estimation

4.3.5.1 The Kalman Filter

4.3.5.2 The Extended Kalman Filter

4.3.5.3 The Invariant Extended Kalman Filter

4.3.5.4 The Two-Frames Group In-EKF

4.4 Tracking Algorithm Design

4.4.1 Software Architecture

4.5 Implementation

4.5.1 Calibration Procedure

4.5.2 Graphical User Interface

4.6 Experimental Testing Setup

4.6.1 Physical Configuration

4.6.2 Reference System and Ground Truth

4.6.2.1 Spatial and Temporal Alignment

How the system's coordinate frame was aligned with OptiTRack - will camera be in their optimised position versus OptiTrack's ad hoc arrangement? How the two data streams would be temporarily aligned

4.6.3 Test Trajectories

4.6.4 Data Acquisition and Processing Pipeline

flow chart

4.7 Results and Discussion

4.7.1 Calibration Accuracy

4.7.2 Static Localisation Performance

4.7.3 Dynamic Tracking Performance

4.7.4 Filter Comparison

4.7.5 Real-time Performance

4.7.6 System Limitations and Failure Analysis

Optimal Camera Placement 5

5.1 Literature Review

5.1.1 Motivation for Optimal Placement

5.1.2 Optimisation Approaches for Sensor Placement

5.1.2.1 Optimisation Objectives

5.1.3 Modelling the Problem

5.1.4 Summary

5.2 Theoretical Framework

5.2.1 Genetic Algorithm Terminology

5.2.2 Encoding Schemes

5.2.3 Selection Methods

5.2.4 Crossover Techniques

5.2.5 Convergence Analysis

5.3 Methodology

5.3.1 Algorithm Configuration

5.3.1.1 Parameter Selection

5.3.1.2 Initial Population

5.3.1.3 Tournament Selection

5.3.1.4 Double Point Crossover

5.3.1.5 Mutation

5.3.1.6 Elitism

5.3.2 Cost Function

5.3.2.1 Resolution Uncertainty Cost Function

5.3.2.2 Dynamic Occlusion Cost Function

5.3.2.3 Combined Cost Function

5.3.3 Termination Criteria

5.4 Results and Discussion

5.4.1 Convergence Behaviour

5.4.2 Cost Function Evaluation

5.4.3 Parameter Sensitivity Analysis

5.4.4 Optimal Placement Configurations

Different Modalities Random Placement, ad hoc placement

5.4.5 System Limitations

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Conclusion and Future Work 6

6.1 Summary of Contributions

6.2 Key Findings

6.3 Limitations

includes scope constraints outlined in intro

6.4 Recommendations for Future Work

closed-loop robotic control Benchmark GA against other metaheuristic like PSO or SA to see if better solutions existan assess convergence performance